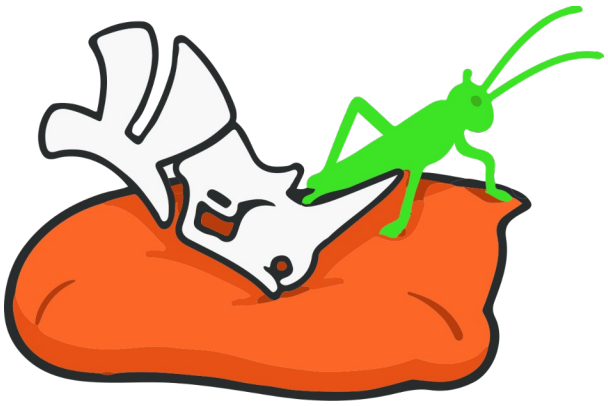


Cushion

Soft robotics simulation tools for Grasshopper3D

Cushion

noun

A soft pad or pillow, typically placed on a sofa, that makes a rigid structure comfortable to work with.

Also: an intermediate layer that softens the interface between two hard things.

Cushion is the soft layer between Rhino and SOFA.

CUSHION V0.1 -- Soft robotics simulation tools for Grasshopper3D

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See the LICENSE file in the repository for the full text.

Cushion interfaces with SOFA (Simulation Open Framework Architecture, www.sofa-framework.org) and its SoftRobots plugin, which are distributed under their own licenses and must be installed separately.

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Abstract

Cushion is a plugin for simulating soft robots directly within Rhino and Grasshopper3D. It connects the parametric design environment to SOFA (Simulation Open Framework Architecture) and its SoftRobots plugin: deformable bodies, pneumatic cavities, tendons, stiffened regions, and rigid interaction objects are defined as native Grasshopper components. Cushion meshes the geometry, writes a complete finite-element scene, launches the simulation, and returns trajectories, actuation forces, and deformed shapes to the canvas -- so design iteration and physical simulation live in the same definition.

The workflow grew out of the Monolithic Unit research pipeline (Exley, Nardin, Trunin, Cafiso and Beccai, 2025), in which soft pneumatic actuators designed parametrically in Grasshopper were simulated in SOFA for actuation and sensor-placement studies. Cushion generalizes that pipeline: any closed solid can become a simulated soft body, with materials, actuation, and boundary conditions assigned per component, and results flowing back for downstream design logic -- workspace analysis, grasp evaluation, or optimization loops.

V0.1 -- Initial release. Pneumatic and tendon actuation, local stiffening regions, rigid targets and contact, scene preview, trajectory monitoring, tendon telemetry, and deformed-shape playback. Requires the freely available SOFA binaries (v24.06).

Dependencies & Installation

- Rhino 8 with Grasshopper.

- SOFA v24.06 binaries with the SofaPython3, SoftRobots, STLIB and SofaValidation plugins (all included in the official release bundle from www.sofa-framework.org/download). Set the SOFA_ROOT environment variable to the installation folder, or wire the path into SofaRun.

- Python 3.10 with numpy, for SOFA's embedded interpreter (see the installation guide for per-platform steps; this is the most common setup issue).

- Gmsh (Python package), installed once into Rhino's own Python: run "<rhino-python>" -m pip install gmsh (the rhino-python path is in the installation guide). Do not rely on automatic install.

Installing Cushion

- From the Rhino Package Manager: run the _PackageManager command, search for Cushion, install, restart Rhino. The Cushion tab appears in Grasshopper.

- Or manually from food4Rhino: download the .gha file, then in Grasshopper open File > Special Folders > Components Folder, copy the .gha there. On Windows, right-click the downloaded file > Properties > Unblock before copying. Restart Rhino.

Links

- Rhino 8: www.rhino3d.com
- SOFA v24.06 binaries: www.sofa-framework.org/download
- Python 3.10 (python.org installer): www.python.org/downloads/release/python-31011
- Cushion repository & issues: [repository URL]
- Manual, examples, install guides: [repository URL]/docs

Verify the SOFA side before using Cushion:

<SOFA>/bin/runSofa -l SofaPython3 -g batch -n 1

Units & Conventions

Millimeters, kilopascals, kilograms, seconds. Gravity -9810 mm/s2 in world Z. Geometry passes from Rhino to SOFA 1:1 -- no transform, no unit conversion. The Rhino document must be in millimeters.

Overview of Tools

[toolbar screenshot placeholder -- replace with a capture of the Cushion tab]

Define

 **SofaBody --** The deformable finite-element body.

 **SofaCavity --** A pneumatic chamber.

 **SofaTendon --** A cable/tendon routed through the body (SOFA CableConstraint).

 **SofaTendons --** Multiple tendons from one component.

 **SofaRegion --** A locally stiffened sub-region (membrane, embedded sensor, waveguide).

 **SofaTarget --** A rigid interaction object: something to grasp, poke, or rest on.

Actuate

 **PressureProfile --** Actuation keyframes for one channel: linear interpolation between (time, value) pairs, resampled into the scene's actuation schedule.

 **SofaProfiles --** Multiple profiles from one component, via trees: one values branch per channel; a single times branch is shared by all.

 **SofaProtocol --** Keyframed rigid motion of a whole body over time -- a pick-and-place move: carry a gripper down, hold, and lift.

Simulate

 **SofaScene --** Assembles everything into a runnable SOFA project folder: meshes the bodies, writes the scene description, actuation schedule, and solver configuration.

 **SofaRun --** Launches the SOFA simulation: interactive GUI (press Animate) or headless batch that blocks until finished.

 **SofaPreview --** Rebuilds the written scene as Rhino geometry: the actual simulation tet mesh, cavities, regions, tendon paths, targets, monitor points, and fixation boxes.

Results

 **SofaResults --** Reads simulation results back into Grasshopper: monitored point trajectories, actuation history, and tendon telemetry.

 **SofaShape --** Rebuilds the deformed body surface at any exported frame.

Component Reference



SofaBody

Define

The deformable finite-element body. Provide a closed solid (Brep preferred) or an existing mesh file; Cushion tet-meshes it via Gmsh and assigns the material properties. All actuation, fixation, monitoring, and display elements attach to a body.

Inputs

Name	Type	Access	Description
geo	Geometry	item	Closed solid. Breps are exported as STEP for exact meshing; meshes as STL.
volumeFile	Text	item	Alternative to geo: .vtk used as-is (pre-meshed tets); .step/.stl is meshed by Gmsh.
name	Text	item	Node name, unique per body (default actuator1). SofaResults/SofaShape reference it.
E	Number	item	Young's modulus, kPa.
nu	Number	item	Poisson ratio (default 0.4).
mass	Number	item	Total mass, kg.
meshSize	Number	item	Gmsh mesh size factor; smaller = finer tets.
gmshOpts	Text	list	Extra Gmsh options as key = value lines (defaults match the reference pipeline), e.g. algo2d = MeshAdapt, algo3d = Delaunay, smoothing = 1, sizeFactor = 0.5.
cavities	Cavity	list	From SofaCavity.
tendons	Tendon	list	From SofaTendon / SofaTendons.
regions	Region	list	From SofaRegion.
fixations	Box	list	Mesh nodes inside each box are clamped to their rest position.
monitorPts	Point	list	Tracked points, snapped to closest mesh nodes; required for trajectory output.
collisionGeo	Geometry	item	Collision surface; wire only when contact matters (costs compute).
collisionFile	Text	item	Alternative: existing collision .stl path.
visualGeo	Geometry	list	Display-only surfaces, deformed with the body but never simulated.
visualFiles	Text	list	Display-only .stl paths, used as-is.
visualFaces	Integer	item	Decimate visualGeo to this face count on export (0 = off).
protocol	Protocol	item	Keyframed rigid motion from SofaProtocol; needs a fixation box (the motion drags the body by its clamps).
exportShape	Integer	item	Export the deformed volume mesh every N steps for SofaShape playback (0 = off).

Outputs

Name			Description
body			Body reference, feed into SofaScene.

Notes: Monitor points also record force: SofaResults returns the nodal force (mN) at each, so a point on a contact pad reads out the grip reaction.



SofaCavity

Define

A pneumatic chamber. The closed surface becomes a SOFA SurfacePressureConstraint driven by the actuation channel; positive pressure inflates, negative deflates.

Inputs

Name	Type	Access	Description
geo	Geometry	item	Closed surface fully inside the body.
meshFile	Text	item	Alternative: existing closed .stl.
channel	Integer	item	Actuation channel (column of the schedule) driving this cavity.
name	Text	item	Default cavity_<channel>.

Outputs

Name			Description
cavity			Cavity reference, feed into SofaBody.

Notes: Pressures are commanded in kPa; internal solver scaling is handled automatically.

SofaTendon

Define

A cable/tendon routed through the body (SOFA CableConstraint). Driven in displacement (mm) or force (mN) through its actuation channel.

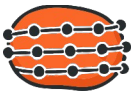
Inputs

Name	Type	Access	Description
pts	Point	list	Ordered routing points, all inside the body (e.g. Divide Curve). First point = motor side.
ptsFile	Text	item	Alternative: JSON file of [[x,y,z], ...] points.
channel	Integer	item	Actuation channel.
name	Text	item	Default cable_<channel>.
valueType	Text	item	displacement (mm, default) or force (mN).
pullPoint	Point	item	Optional external anchor (motor/pulley) the cable exits toward.

Outputs

Name			Description
tendon			Tendon reference, feed into SofaBody.
pathPts			Routing points for viewport preview.

Notes: Driving several coupled cables simultaneously: use force mode. Every run also records displacement and force per cable (see SofaResults).



SofaTendons

Define

Multiple tendons from one component. Wire a tree of points (one branch per tendon); channels default to 1..N in branch order.

Inputs

Name	Type	Access	Description
pts	Point	tree	One branch per tendon (an un-flattened Divide Curve over a list of curves).
channels	Integer	list	Channel per branch (optional, default 1..N).
valueType	Text	item	displacement or force, applied to all tendons.
pullPoints	Point	list	Optional anchor per tendon, matched by index.

Outputs

Name			Description
tendons			List of tendon references, feed into SofaBody (no Merge needed).
info			Summary of created tendons.



SofaRegion

Define

A locally stiffened sub-region (membrane, embedded sensor, waveguide). The closed mesh selects body elements and overlays additional stiffness on them; the region is never a separate meshed part.

Inputs

Name	Type	Access	Description
geo	Geometry	item	Closed mesh enclosing the elements to stiffen.
surface	Geometry	item	Alternative: base surface for the auto-shell mode.
thickness	Number	item	Shell offset (mm) used with surface: builds a closed hollow shell (a membrane with no inlet, impossible as a Brep).
meshFile	Text	item	Alternative: existing closed .stl.
name	Text	item	
E	Number	item	Overlay modulus, kPa, added on top of the body material.
nu	Number	item	Default 0.4.
repo	Text	item	Cushion folder path (auto-shell mode only; not needed with the installed plugin).

Outputs

Name			Description
region			Region reference, feed into SofaBody.
shell			Generated shell mesh (auto-shell mode preview).



SofaTarget

Define

A rigid interaction object: something to grasp, poke, or rest on. Falls under gravity and responds to contact with any body that has a collision surface.

Inputs

Name	Type	Access	Description
geo	Geometry	item	Closed surface, placed in world coordinates.
meshFile	Text	item	Alternative: existing closed .stl.
name	Text	item	Default target1.
mass	Number	item	kg (default 0.01).

Outputs

Name			Description
target			Target reference, feed into SofaScene.

Notes: Contact requires collisionGeo on the body, and usually a floor (SofaScene.floorZ). Rotational inertia is approximated in this release.



PressureProfile

Actuate

Actuation keyframes for one channel: linear interpolation between (time, value) pairs, resampled into the scene's actuation schedule. Values are kPa for cavities, mm or mN for tendons.

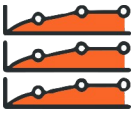
Inputs

Name	Type	Access	Description
channel	Integer	item	Channel this profile drives.
times	Number	list	Keyframe times, seconds.
values	Number	list	Keyframe values.

Outputs

Name			Description
profile			Profile reference, feed into SofaScene.

Notes: Values hold at the last keyframe: add an explicit return to 0 if the channel should release.



SofaProfiles

Actuate

Multiple profiles from one component, via trees: one values branch per channel; a single times branch is shared by all.

Inputs

Name	Type	Access	Description
channels	Integer	list	Channel per branch (optional, default 1..N).
times	Number	tree	Keyframe times per channel; one branch = shared timeline.
values	Number	tree	Keyframe values, one branch per channel.

Outputs

Name			Description
profiles			List of profile references, feed into SofaScene.
info			Summary.



SofaProtocol

Actuate

Keyframed rigid motion of a whole body over time -- a pick-and-place move: carry a gripper down, hold, and lift. The motion is applied through the body's rest positions, so the body is dragged by its fixations; a body without a fixation box will not follow.

Inputs

Name	Type	Access	Description
times	Number	list	Keyframe times, seconds.
offsets	Vector	list	Body offset from its start position at each keyframe (first is usually 0,0,0). Linear interpolation between keyframes.
name	Text	item	Optional label.

Outputs

Name			Description
protocol			Protocol reference, feed into SofaBody.

Notes: Time the actuation profiles to match the motion -- e.g. inflate the cavities while the offset holds at the object.



SofaScene

Simulate

Assembles everything into a runnable SOFA project folder: meshes the bodies, writes the scene description, actuation schedule, and solver configuration. The folder is self-contained and can also be run outside Grasshopper.

Inputs

Name	Type	Access	Description
bodies	Body	list	From SofaBody.
profiles	Profile	list	From PressureProfile / SofaProfiles.
targets	Target	list	From SofaTarget (optional).
camPos	Point	item	SOFA camera position (optional).
camTarget	Point	item	Camera look-at point (optional, needs camPos).
viewFile	Text	item	A saved SOFA .view file to use instead (Save View in the SOFA GUI).
projDir	Text	item	Output folder, local disk. One folder per design.
repo	Text	item	Cushion folder path (not needed with the installed plugin).
dt	Number	item	Timestep, s (default 0.01).
gravityZ	Number	item	mm/s2; default -9810, use 0 while testing.
floorZ	Number	item	Adds a static floor plane at this height (empty = none).
floorSize	Number	item	Floor edge length, mm (default 400).
contactRules	Text	item	Advanced contact pairing rules (empty = default friction contact).
write	Boolean	item	Toggle True to (re)write the project.

Outputs

Name			Description
scenePath			Path to the written scene; feed into SofaRun / SofaResults / SofaShape / SofaPreview.
info			What was written; read this first when something is off.

Notes: Units: mm / kPa / kg / s throughout. Rhino document units must be millimeters.

SofaRun

Simulate

Launches the SOFA simulation: interactive GUI (press Animate) or headless batch that blocks until finished.

Inputs

Name	Type	Access	Description
scenePath	Text	item	From SofaScene.
sofaRoot	Text	item	SOFA installation folder (or set the SOFA_ROOT environment variable).
repo	Text	item	Cushion folder path (not needed with the installed plugin).
gui	Boolean	item	True: SOFA window; False: headless batch.
winRect	Integer	list	Optional x, y, width, height to place the SOFA window (Windows).
run	Boolean	item	Toggle True to launch.

Outputs

Name			Description
status			Launch/exit status; the run log is written next to the scene.

SofaPreview

Simulate

Rebuilds the written scene as Rhino geometry: the actual simulation tet mesh, cavities, regions, tendon paths, targets, monitor points, and fixation boxes. Wire outputs through Custom Preview components for per-role colors.

Inputs

Name	Type	Access	Description
scenePath	Text	item	From SofaScene.
repo	Text	item	Cushion folder path (not needed with the installed plugin).
refresh	Boolean	item	Toggle to re-read.

Outputs

Name			Description
volumes			Simulation mesh surfaces (meshing QA).
cavities			Cavity meshes.
regions			Region meshes.
targets			Target meshes.
tendonCrvs			Cable routing polylines (pull point included).
tendonChs			Channel number per tendon.
monitorPts			Tracked points.
fixBoxes			Fixation boxes.
info			Counts and any skipped mesh files.

SofaResults

Results

Reads simulation results back into Grasshopper: monitored point trajectories, actuation history, and tendon telemetry.

Inputs

Name	Type	Access	Description
scenePath	Text	item	From SofaScene.
bodyName	Text	item	Body whose monitor to read (SofaBody name).
repo	Text	item	Cushion folder path (not needed with the installed plugin).
refresh	Boolean	item	Toggle to re-read.

Outputs

Name			Description
info			What was found; read this first.
times			One time per recorded frame.
points			Tree: branch per frame, items = tracked point positions.
lastPts			Final frame (quick preview).
forces			Tree of vectors: nodal force (mN) at the monitor points, branch per frame -- the grip reaction where a point sits on a contact pad.
pTimes			Actuation sample times.
pressures			Tree: branch per channel, commanded/measured values.
cableNames			Cable names (when tendons exist).
cableDisp			Tree: displacement per cable, mm.
cableForce			Tree: force per cable, mN.
contactPairs			Names of the body/target contact pairs (when targets and collision exist).
contactCount			Tree: number of contacts per step, branch per pair.
contactTimes			Tree: step times, branch per pair.
contactPts			Tree: contact locations on the body surface, branch per pair and frame (contact geometry, not force).

Notes: Typical recipes: points + Tree Branch + slider = scrub the motion; Flip Matrix + Polyline = trajectory curves; pressures + Quick Graph = actuation history; contactPts at a frame = the grasp contact map.



SofaShape

Results

Rebuilds the deformed body surface at any exported frame. Requires exportShape set on SofaBody when the scene was written.

Inputs

Name	Type	Access	Description
scenePath	Text	item	From SofaScene.
bodyName	Text	item	Body to read.
frame	Integer	item	Frame index; empty = last. Wire a slider 0..nFrames-1 to scrub.
repo	Text	item	Cushion folder path (not needed with the installed plugin).
refresh	Boolean	item	Toggle to re-read.

Outputs

Name			Description
mesh			Deformed surface mesh at the frame.
nFrames			Frames available.
time			Simulation time of the frame, s.
info			Status.

Workflow Example: Tendon-Driven Continuum Robot

A cylindrical continuum robot with three tendons at 120 degrees, built entirely from native Grasshopper geometry, actuated with a rotating pull direction, and post-processed into a workspace boundary.

1. Body: Circle (r 12) > Extrude (z 100) > Cap Holes > SofaBody (E 100 kPa, mass 0.03 kg).

2. Tendons: three lines at radius 8, Divide Curve > SofaTendons (tree input), valueType force.

3. Fixation: box around the base; monitor points along the axis.

4. Profiles: 13 keyframes per channel sweeping Tmax * max(0, cos(30 deg * k - phi)) -- the resultant pull direction rotates 360 degrees.

5. SofaScene (gravity 0) > SofaPreview to verify > SofaRun.

6. SofaResults: tip positions per frame > convex hull = workspace bounds; cableForce/cableDisp confirm the commanded lobes.

[screenshot placeholders: canvas, SOFA window, workspace hull]